

Newts and salamanders

PHYLUM	Chordata
CLASS	Amphibia
ORDER	Caudata
FAMILIES	9
SPECIES	707

Newts and salamanders have a slender body with a long tail and, typically, 4 legs of similar size. Of the 3 main groups of living amphibians, they most closely resemble the animals from which

all amphibians are descended. They also have some of the most complex and varied life histories. Some salamanders live their entire lives in water, while others are wholly terrestrial. Newts spend most of their lives on land but return to water to breed—other than this, there is no scientific distinction between salamanders and newts. In all but the most primitive species, fertilization is internal (rather than external, as in most frogs and toads). The male does not have a penis but instead passes capsules of sperm to the female during mating. The larvae of newts and salamanders are carnivorous, and have a long, slender body, a deep, finlike tail, and large, feathery external gills. Newts and salamanders are found in damp places and are largely confined to the Northern Hemisphere.

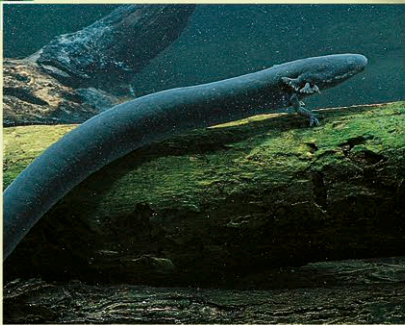
Anatomy

Newts and salamanders differ from frogs and toads in that they retain their tail into adulthood. They also have a relatively small head and small eyes, smell being the most important sense for finding food and social interaction. Most newts and salamanders have 4 legs (with 4 digits on the front limbs and 5 on the back), although some aquatic salamanders have no hind legs and only small front legs. Many salamanders have distinct moisture-maintaining “costal” grooves that run around the body. Newts and salamanders vary in the way that they take in oxygen. Some are able to breathe through their smooth, moist skin, while many of those species that spend all their life in water retain the large, feathery external gills characteristic of the larval stage.



LUNGLESS SALAMANDERS
The plethodontids are a group of typically wholly terrestrial salamanders. They have no lungs but instead breathe through their skin and their mouth and throat, which have a rich blood supply.

SIRENS
The sirens, of which there are 4 species, spend their entire lives in water. In addition to having lungs, like some other aquatic salamanders, they take in oxygen through large, feathery external gills on the side of their head. In periods of drought, they can survive for several weeks or months, encased deep inside a cocoon of dried mud. During this time, they feed on their large fat reserves.



SALAMANDER HUNTING
Salamanders have wider mouths than newts and can catch larger prey like this earthworm. Many have a long tongue that flicks out then opens up to engulf the prey.

Feeding

All newts and salamanders are carnivorous and feed on living prey. They find their prey using a combination of smell and vision; in addition, some aquatic species are very sensitive to water currents made by moving animals. Since they are rarely very active, newts and salamanders do not have a high energy requirement and do not need to feed very often, especially if they have eaten a large amount. When food is abundant, they accumulate fat deposits that enable them to survive periods when it is too dry or too cold for them to feed. The larvae are also carnivorous, eating a wide variety of aquatic invertebrates. In some species, faster-growing larvae become cannibalistic, eating smaller larvae of their own species.



COURTSHIP
The courtship of newts and salamanders is varied and often elaborate. Courtship precedes and accompanies the transfer of sperm from male to female in capsules known as spermatophores. The male stimulates the female by secreting compounds known as pheromones. These are usually delivered via the nose but in some species they are passed directly through the skin. European newts—such as these Alpine newts (here the male is on the left)—use visual displays rather than contact to stimulate the female, but in many salamanders the male holds the female in a mating embrace known as amplexus while he stimulates her.